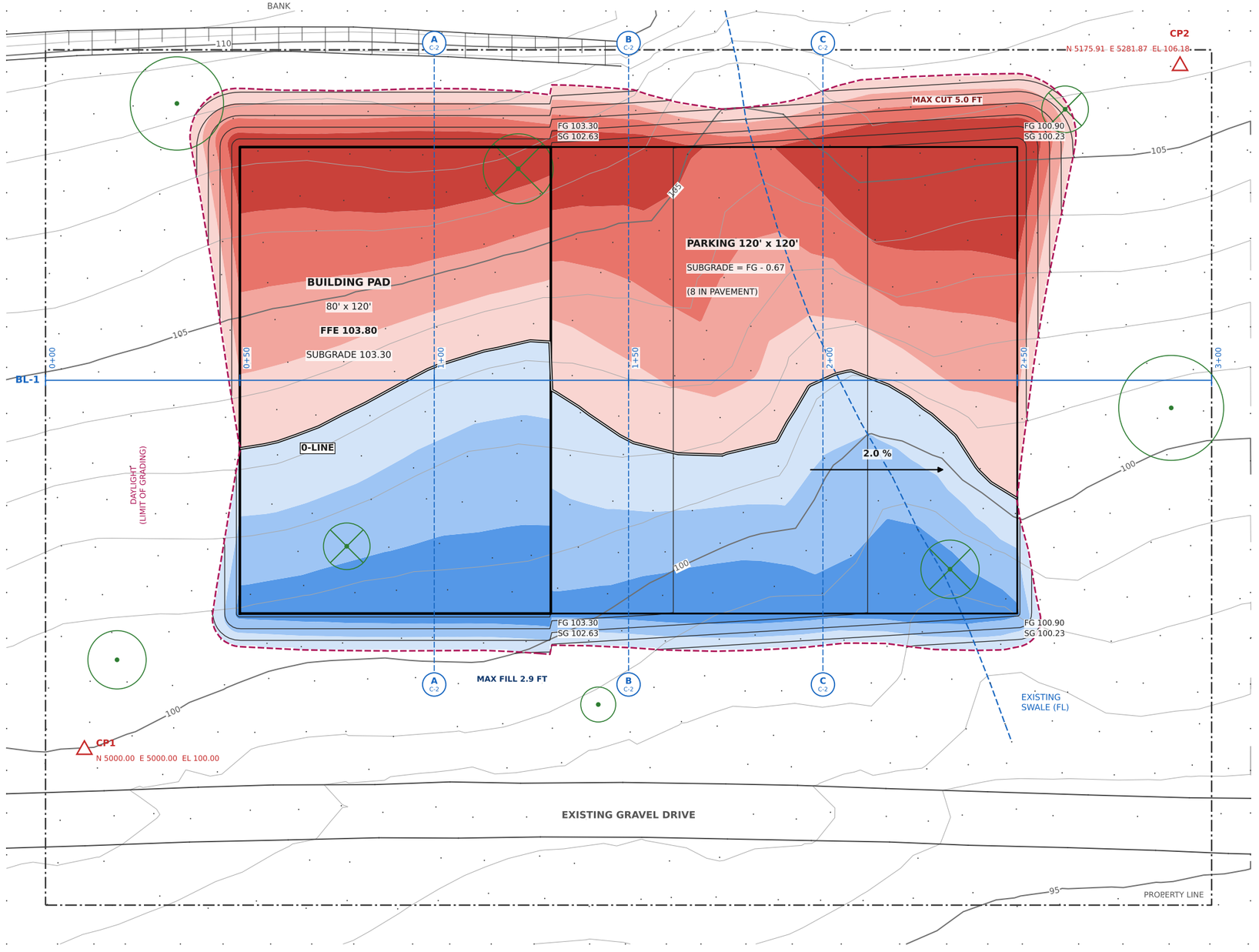


CUT / FILL DEPTH

existing ground to subgrade

- CUT 3 - 5 FT — 3,909 SF
- CUT 2 - 3 FT — 4,602 SF
- CUT 1 - 2 FT — 4,785 SF
- CUT 0 - 1 FT — 4,745 SF
- FILL 0 - 1 FT — 4,449 SF
- FILL 1 - 2 FT — 4,847 SF
- FILL 2 - 3 FT — 3,220 SF

LINEWORK

- EXISTING CONTOUR 1 FT
- EXISTING 5 FT
- PROPOSED SUBGRADE CONTOUR 1 FT
- DAYLIGHT LINE
- 0-LINE
- PAD / PARKING EDGE
- SWALE FLOWLINE
- EDGE OF GRAVEL
- BASELINE BL-1
- SURVEY SHOT
- CONTROL POINT
- TREE / TO REMOVE

N

SCALE 1" = 30' (FEET)

0 30 60 90

EARTHWORK VOLUMES

TIN to TIN, existing ground to subgrade

ITEM	CY
Cut	1,302
Fill	632
Topsoil strip 0.50 ft (separate item)	566
Cut less topsoil in cut areas	968
Fill plus strip in fill areas	864
Fill after shrink 15 % (/ 0.85)	1,016
NET IMPORT	**48**

Disturbed area 30,557 SF (18,042 cut, 12,516 fill).

METHOD CHECK (CY)

METHOD	CUT	FILL	CUT vs TIN	FILL vs TIN
TIN to TIN	1,302	632	main	
Grid 25 ft	1,277	628	-1.9 %	-0.6 %
End areas	1,323	625	+1.6 %	-1.0 %

DESIGN DATA

FFE 103.80, pad subgrade 103.30 (FFE - 0.50 slab and base)

Parking FG 103.30 at the building, 2.0 % down to the east

Parking subgrade = FG - 0.67 (8 in pavement): 102.63 to 100.23

Side slopes 3:1 in cut and fill to daylight

Topsoil strip 0.50 ft, shrink 15 % (assumed)

NOTES

1. Sample survey data prepared for this portfolio: 667 total station points (PNEZD): TOPO, EP, FL, TB, TOE, TREE and control CP1, CP2.
2. Local assumed grid and datum: CP1 = N 5000.000, E 5000.000, EL 100.000. US survey ft.
3. Existing TIN: shots plus EP, FL, TB, TOE breaklines.
4. Volumes computed in Python: TIN to TIN at 0.5 ft; grid and end areas in Earthworks_Volumes.xlsx.
5. 4 trees inside the grading limits to be removed (marked X); clearing not included.
6. Footings, utility trenches, undercut, rock, driveway connection: not included. See Notes & Queries.

1 / C-1

CUT / FILL PLAN

SCALE: 1" = 30'-0"

Colours: depth from existing ground to subgrade (pad, parking and 3:1 slopes).

Contours: existing ground from the survey TIN (grey); proposed subgrade (black).

Sections A, B, C on sheet C-2. 25 ft volume grid with depths: layer C-VOL-GRID in the DXF.

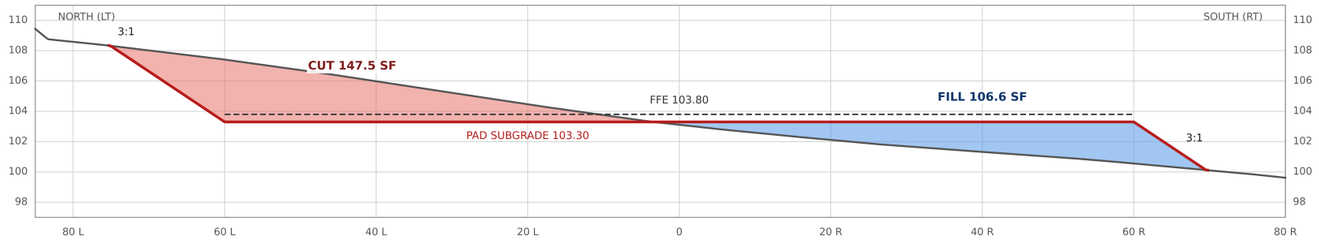
SAMPLE SURVEY DATA PREPARED FOR THIS PORTFOLIO

Earthworks_Sample.dxf | Earthworks_Volumes.xlsx | Survey_Points_PNEZD.csv

PROJECT	**COMMERCIAL SITE - PAD AND PARKING**
	PORTFOLIO SAMPLE
SHEET	**CUT / FILL PLAN AND VOLUMES**
DRAWN	**SERGEI GRISHUTOV**
SCALE	1" = 30' @ 11x17
DATE	NOV 2026

NOT FOR CONSTRUCTION

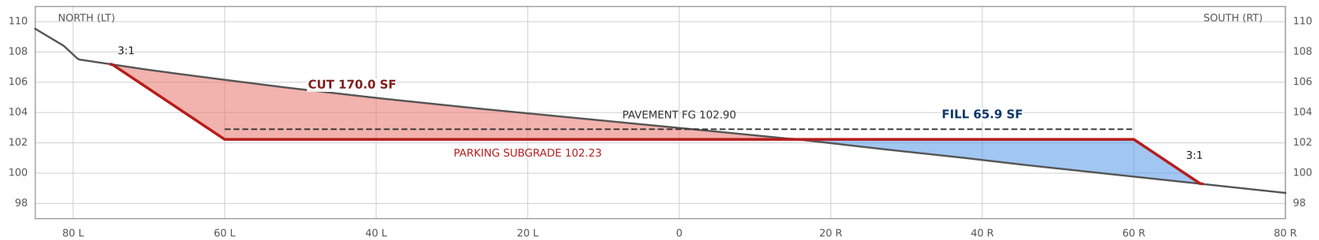
C-1



A
C-2

SECTION A - STA 1+00 ON BL-1 (E 5090), LOOKING EAST

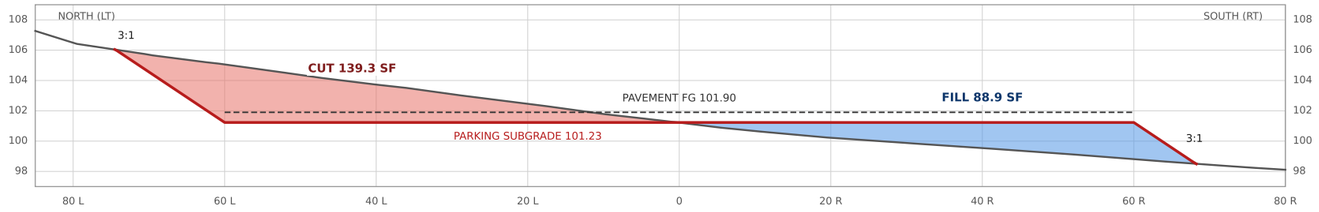
H: 1" = 20' V: 1" = 10' (vertical exaggeration 2x)



B
C-2

SECTION B - STA 1+50 ON BL-1 (E 5140), LOOKING EAST

H: 1" = 20' V: 1" = 10' (vertical exaggeration 2x)



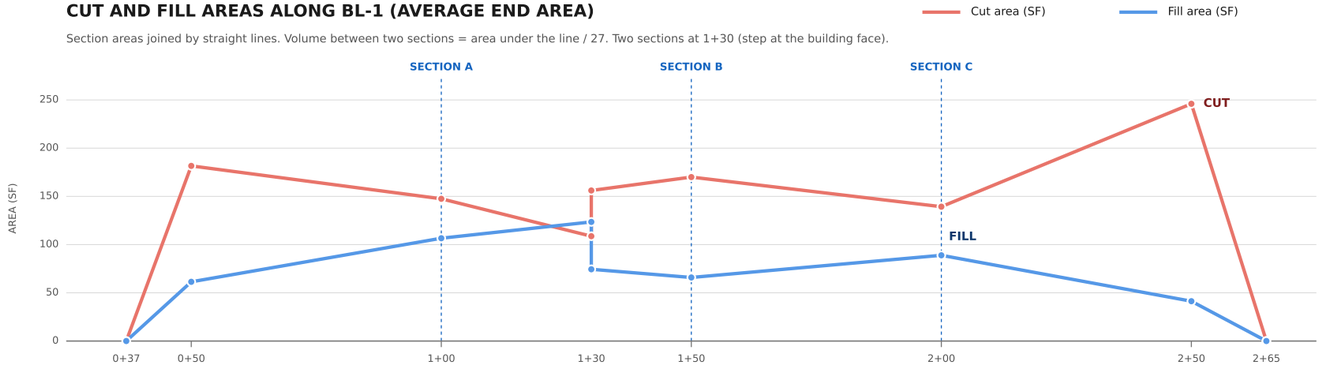
C
C-2

SECTION C - STA 2+00 ON BL-1 (E 5190), LOOKING EAST

H: 1" = 20' V: 1" = 10' (vertical exaggeration 2x)

CUT AND FILL AREAS ALONG BL-1 (AVERAGE END AREA)

Section areas joined by straight lines. Volume between two sections = area under the line / 27. Two sections at 1+30 (step at the building face).



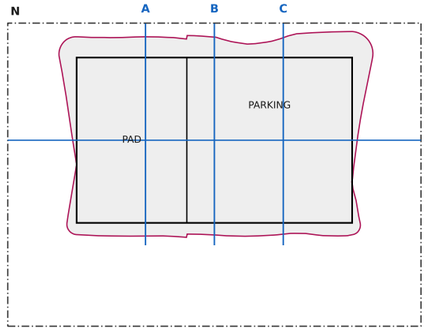
LEGEND

- EXISTING GROUND (TIN)
- PROPOSED SUBGRADE
- FINISHED GRADE / FFE
- CUT AREA
- FILL AREA

SECTION NOTES

1. Sections are cut from the existing TIN and the design surface at the BL-1 stations.
2. Areas between existing ground and subgrade, used in the average end area table.
3. All 9 sections are drawn at true scale in Earthworks_Sample.dxf (layers C-SECT-*).
4. Offsets from BL-1: L = north, R = south.
5. Topsoil strip and shrink are applied to the TIN result on sheet C-1, not per section.

KEY PLAN



Not to scale (about 1" = 110'). Shaded: inside the daylight line.

AVERAGE END AREA - BL-1

V = (A1 + A2) / 2 x L / 27

STA	CUT SF	FILL SF	CUT CY	FILL CY
0+37	0.0	0.0		
0+50	181.6	61.4	43.7	14.8
1+00	147.5	106.6	304.7	155.6
1+30 pad	108.7	123.5	142.3	127.8
1+30 park	156.1	74.4	0.0	0.0
1+50	170.0	65.9	120.8	52.0
2+00	139.3	88.9	286.4	143.3
2+50	246.0	41.3	356.8	120.6
2+65	0.1	0.0	68.4	11.5
TOTAL			**1,323.1**	**625.5**

Two sections at 1+30: the subgrade steps 0.67 ft at the building face (pad side / parking side).

METHOD CHECK (CY)

METHOD	CUT	FILL	CUT vs TIN	FILL vs TIN
TIN to TIN	1,302	632	main	
Grid 25 ft	1,277	628	-1.9 %	-0.6 %
End areas	1,323	625	+1.6 %	-1.0 %

WHY THE METHODS DIFFER

Grid: 25 ft cells straighten the swale and the change of slope at the pad edges. Cells cut by the daylight line use the area inside it.

End areas: a straight-line change is assumed between sections. The slopes taper out faster at both ends (high); the swale crosses BL-1 at an angle under the parking (fill low between 2+00 and 2+50).

All three agree within 2 %.

PROJECT	**COMMERCIAL SITE - PAD AND PARKING**
	PORTFOLIO SAMPLE
SHEET	**SECTIONS AND END-AREA VOLUMES**
DRAWN	**SERGEI GRISHUTOV**
SCALE	AS NOTED @ 11x17
DATE	NOV 2026

NOT FOR CONSTRUCTION

C-2